

PRINCIPAL INDUSTRIES ATTRIBUTED FOR WATER POLLUTION

Introduction

Water, because of its properties as a solvent and its capacity to transport particles, incorporates in itself various impurities that characterize the water quality. Water quality is a result of natural phenomena and the acts of human beings. Generally, it can be said that water quality is a function of land use in the catchment area. This is due to the following factors:

- a) Natural conditions: even with the catchment area preserved in its natural condition, the surface water quality is affected by run off and infiltration resulting from rainfall. The impact of these is dependent on the contact of the water with particles, substances and impurities in the soil. Therefore, the incorporation of suspended solids (e.g. soil particles) or dissolved solids (e.g. ions originating from the dissolution of rocks) occurs even when the catchment area is totally preserved in its natural condition (e.g. occupation of the land with woods and forests). In this case, the soil protection and composition have a great influence.
- b) Interference of human beings: the interference of man manifests itself either in a concentrated form, such as in the discharge of domestic or industrial wastewater, or in a diffused form, such as in the application of fertilizers or pesticides onto the soil. Both contribute to the introduction of compounds into the water, thus affecting its quality. Therefore, the form in which human beings use and occupy the land has a direct implication in the water quality.

Uses of water

The main water uses are:

- domestic supply
- industrial supply
- irrigation
- animal supply
- preservation of aquatic life
- recreation and leisure
- breeding of aquatic species
- generation of electricity
- navigation
- landscape harmony

➤ dilution and transport of wastes

In general terms, only the first two uses (domestic supply and industrial supply) are frequently associated with a prior water treatment, in view of their more demanding quality requirements.

There is a direct relation between water use and its required quality. In the above list, the most demanding use can be considered domestic water supply, which requires the satisfaction of various quality criteria. Conversely, the less demanding uses are simple dilution and transportation of wastes, which do not have any specific requirements in terms of quality. However, it must be remembered that multiple uses are usually assigned to water bodies, resulting in the necessity of satisfying diverse quality criteria. Such is the case, for example, of reservoirs constructed for water supply, electricity generation, recreation, irrigation and others.

Besides the cycle of water on Earth (hydrological cycle), there are internal cycles, in which water remains in the liquid state, but has its characteristics modified as a result of its use. Figure 1 shows an example of typical routes of water use, composing partial cycles. In these cycles, the water quality is modified at each stage of its journey. The management of these internal cycles is an essential role in environmental engineering, and includes the planning, design, construction and control of the works necessary for the maintenance of the desired water quality as a function of its intended uses. Therefore, the engineer or scientist must know how to ask for and interpret the results of water quality samples in the various points of the cycle.

- Drinking and domestic purposes: Water is necessary for drinking, cooking, bathing, washing clothes, and cleaning households.
- Agriculture: Water is widely used for irrigation, livestock farming, and food production. Agriculture is one of the largest consumers of freshwater worldwide.
- Industrial Uses: Industries use water for manufacturing processes, cooling machinery, cleaning equipment, and generating steam.
- Hydroelectric power generation: Water is used to produce electricity through hydroelectric power plants, providing a renewable source of energy.
- Transportation: Rivers, canals, lakes, and oceans serve as important routes for the transportation of goods and people.
- Recreation and tourism: Water bodies are used for swimming, boating, fishing, and other recreational activities that contribute to tourism and well-being.
- Environmental support: Water sustains aquatic ecosystems, supports biodiversity, and helps maintain ecological balance.
- Public services: Water is used in firefighting, sanitation systems, hospitals, schools, and other public facilities.

ROUTES OF WATER USE AND DISPOSAL

water, wastewater and stormwater

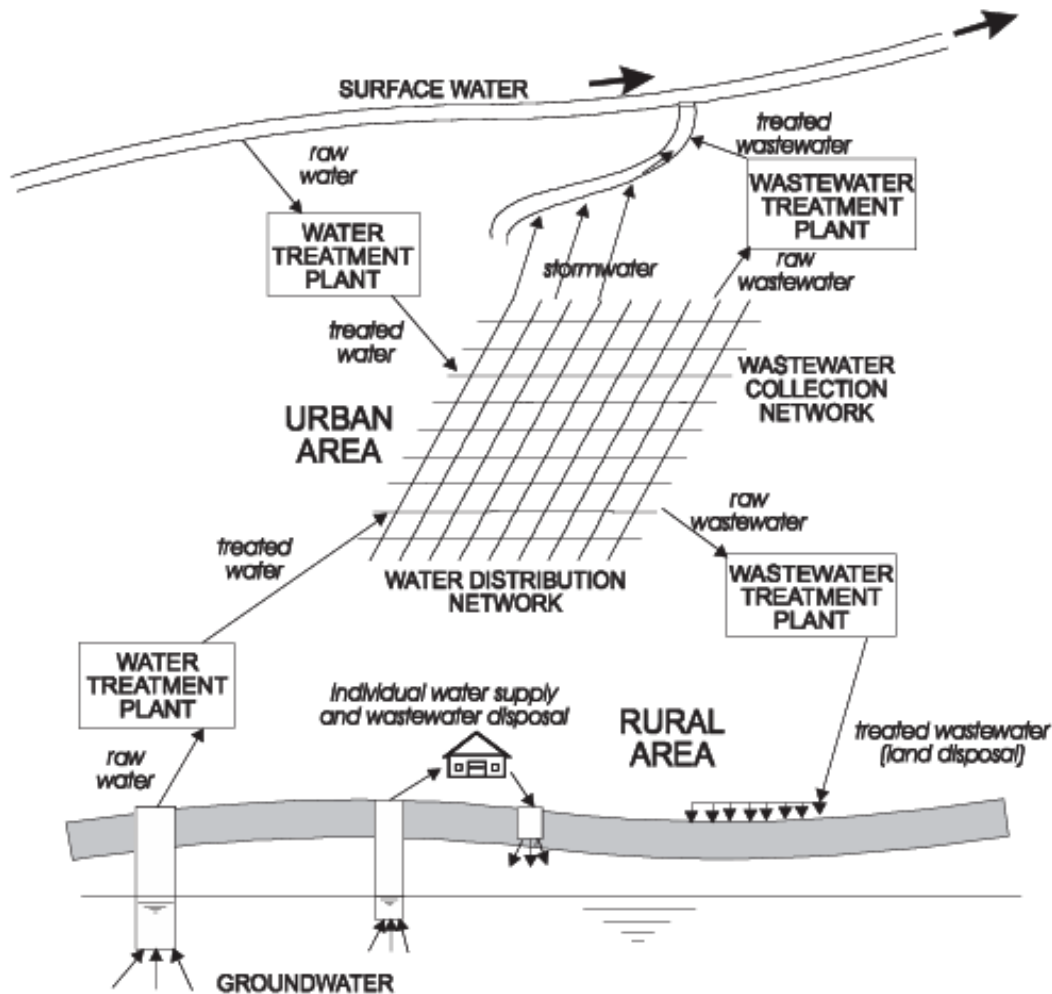


Fig 1: Routes of water use

- Raw-water: Initially, water is abstracted from the river, lake or water table, and has a certain quality.
- Treated water: After abstraction, water undergoes transformations during its treatment to be able to comply with its intended uses (e.g. public or industrial water supply).
- Raw wastewater: The water, after being used, undergoes new transformations in its quality and becomes a liquid waste.
- Treated wastewater: Aiming at removing its main pollutants, wastewater undergoes treatment before being discharged into the receiving body. Wastewater treatment is responsible for the new modification in the quality of the liquid.

- Stormwater: Rain water flows on the ground, incorporates some pollutants, and is collected at stormwater systems before being discharged into the receiving body.
- Receiving body: Stormwater and the effluent from the wastewater treatment plant reach the receiving body where water quality undergoes new modifications, as a result of dilution and self-purification mechanisms.

Water quality requirements

Table 1 presents in a simplified way the association between the main quality requirements and the corresponding water uses. In cases of water bodies with multiple uses, the water quality must comply with the requirements of the various intended uses.

Table 1: Association between water use and quality requirements

General use	Specific use	Required quality
Domestic supply	-	- Free from chemical substances harmful to health - Free from organisms harmful to health - Low aggressiveness and hardness - Aesthetically pleasant (low turbidity, color, taste and odour; absence of macro-organisms)
Industrial supply	Water incorporated into the product (e.g. food, drinks, medicines)	- Free from chemical substances harmful to health - Free from organisms harmful to health - Aesthetically pleasant (low turbidity, color, taste and odour; absence of macro-organisms)
	Water that enters into contact with the product	-Variable with the product
	Water that does not enter into contact with the product (e.g. refrigeration units, boilers)	- Low hardness - Low aggressiveness

Irrigation	Horticulture, products ingested raw or with skin	- Free from chemical substances harmful to health - Free from organisms harmful to health - Non-excessive salinity
	Other plantations	- Free from chemical substances harmful to the soil and plantations - Non-excessive salinity
Animal water supply	-	-Free from chemical substances harmful to animals health - Free from organisms harmful to animals health
Preservation of aquatic life	-	Variable with the environmental requirements of the aquatic species to be preserved
Aquaculture	Animal breeding	-Free from chemical substances harmful to animals, workers and consumers health - Free from organisms harmful to animals, workers and consumers health - Availability of nutrients
	Vegetable growing	- Free from chemical substances toxic to vegetables and consumer - Availability of nutrients
Recreation and leisure	Primary contact (direct contact with the liquid medium– bathing; e.g.: swimming, water-skiing, surfing)	- Free from chemical substances harmful to health - Free from organisms harmful to health - Low levels of suspended solids and oils and grease
	Secondary contact (without direct contact with the liquid medium; e.g.: leisure navigation, fishing, contemplative viewing)	Pleasant appearance

Energy generation	Hydroelectric power plants	Low aggressiveness
	Nuclear or thermoelectric power plants (e.g. cooling towers)	Low hardness
Transport	-	Low presence of coarse material that could be dangerous to vessels

Analysis of water

The water needs to be tested and treated before distribution. The aim of laboratory test is to ensure that portable water conforming to standards is supplied to consumers. The following tests are conducted:

- To determine the quality of raw water
- To determine the treatment processes to be provided
- To ensure that treatment of water is properly done during each phase or stage of treatment
- To examine whether the treated water conforms to standards

Types of tests

The tests conducted to determine the quality of water are:

1. Physical tests: Physical tests indicate the aesthetic quality and performance of various treatment units.
2. Chemical tests: Chemical tests indicate the amount of chemicals present in water and the amount of pollution in it.
3. Bacteriological tests: Bacteriological tests indicate the presence of bacteria producing pollution and whether the water is safe or not for drinking purpose.

1. Physical tests

These evaluate the visible and sensory characteristics of water.

- a) Temperature – affects aquatic life and chemical reactions.
- b) Color – unusual color may indicate contamination.
- c) Odor – unpleasant smells can suggest pollution or bacterial activity.
- d) Turbidity – measures cloudiness caused by suspended particles.

2. Chemical tests

These measure dissolved substances in water.

- a) pH – Indicates whether water is acidic or alkaline.
- b) Total Dissolved Solids (TDS) – Concentration of dissolved minerals and salts.
- c) Hardness – Mainly caused by calcium and magnesium. Temporary hardness is due to carbonate and bicarbonate of calcium and magnesium while permanent hardness is due to chloride and sulfate of calcium and magnesium.
- d) Chloride – The concentration of chloride ions is quite low in natural water systems. It increases in contaminated water. Moreover, high concentration of chloride ions makes the water salty and corrodes pipelines of water.
- e) Nitrate and Nitrite – Excessive amounts can be harmful, especially to infants.
- f) Iron and Manganese – May cause staining and affect taste.
- g) Heavy Metals – Such as lead, arsenic, mercury, and cadmium.

3. Bacteriological tests

Water may contain bacteria. Bacteria are very small organisms which can be seen only under a microscope. Some bacteria are harmful and are called pathogenic bacteria. Some others are harmless and are called non-pathogenic bacteria. Pathogenic bacteria cause typhoid, cholera, dysenteries, etc.

The most common bacteria in the coliform group is the Escherichia coli (E. coli). They are harmless but their presence indicates the possibility that pathogenic bacteria are also present. E. coli germs are discharged in very large numbers with the faeces.

The following tests are done for bacteriological examination of water:

- I. Total count test
- II. E. coli test or coliform test

I. Total Count Test: In this test the sample of water with agar added to it, is placed in an incubator at 20°C for 48 hours or 37°C for 24 hours. The bacteria in the water grow and form colonies which can be seen and counted. The total count should not be greater than 100 per cc of water.

II. E. Coli Test: It is observed that at 37°C, coliforms in lactose medium ferment within 48 hours with formation of gas. The E. coli test is done in three stages:

- a) Presumptive test
- b) Confirmed test
- c) Completed test

a) Presumptive Test: Samples of water are placed in sterile tubes and lactose is added to them. The tubes are incubated for 24 hours at 37°C. If gas is formed, then the test is positive indicating the presence of bacteria. If no gas is formed, it is again examined at the end of 48 hours. If still gas is formed, the test is negative.

(b) Confirmed Test: This test is carried out to see if E. coli group is present in water. The following two methods are adopted.

- i. When a sample of water is positive in the presumptive test, a small portion of the lactose broth is transferred to another fermentation tube containing **green lactose bile**. After 48 hours if gas formation is seen, the test is positive showing that the water is unsafe for drinking.
- ii. A Small quantity of the sample of water is taken on a plate containing **endo- or eosin-methylene-blue agar**. The plates are kept at 37°C for 24 hours. If colonies are seen., then the result is positive. A completed test is then carried out to establish the presence of E. coli group of bacteria.

(c) Completed Test: This test is carried out by adding the sample of water to lactose broth fermentation tubes and agar tubes. The tubes are incubated for 24 to 48 hours at 37°C. If gas is formed after this period, then the result is positive and the water is not safe for drinking. Further tests are conducted to find out the type of bacteria present in the water. If the test is negative, then the water is safe for drinking.

Standard of quality of water

Parameter	Acceptable/Guideline Value	Significance
Color	≤ 15 TCU	Water should be clear and aesthetically acceptable
Odor	Unobjectionable	No unpleasant smell
Turbidity	≤ 5 NTU (preferably < 1 NTU)	Indicates suspended particles
pH	6.5 – 8.5	Measures acidity or alkalinity
Total Dissolved Solids(TDS)	≤ 500 mg/L (desirable)	Indicates dissolved minerals and salts
Total Hardness	≤ 200 mg/L as CaCO_3 (desirable)	Affects taste and scaling
Chloride	≤ 250 mg/L	High levels affect taste
Sulfate	≤ 250 mg/L	Excess may cause laxative effects
Nitrate (NO_3^-)	≤ 50 mg/L	High levels can affect infants' health
Iron	≤ 0.3 mg/L	Prevents staining and taste issues
Manganese	≤ 0.1 mg/L	Prevents staining and discoloration

Lead	≤ 0.01 mg/L	Toxic even at low concentrations
Copper	≤ 2.0 mg/L	High levels affect taste and health
Zinc	≤ 3.0 mg/L	Mainly affects taste
Total Coliform Bacteria	0 per 100 mL	Indicates microbiological safety
E. coli	0 per 100 mL	Must be absent in drinking water

Water pollution

Water pollution is the contamination of water bodies, which has a negative impact on how they can be used. It is usually caused by human activities. Water bodies include lakes, rivers, oceans, aquifers, reservoirs and groundwater. Water pollution results when contaminants mix with these water bodies. Contaminants can come from one of four main sources. These are sewage discharges, industrial activities, agricultural activities, and urban runoff including stormwater. Water pollution may affect either surface water or groundwater. This form of pollution can lead to many problems. One is the degradation of aquatic ecosystems. Another is spreading water-borne diseases when people use polluted water for drinking or irrigation. Water pollution also reduces the ecosystem services such as drinking water provided by the water resource.

Source of water pollution

Water pollutants come from either point sources or dispersed sources.

A point source is a pipe or channel, such as those used for discharge from an industrial facility or a city sewerage system.

A dispersed (or nonpoint) source is a very broad unconfined area from which a variety of pollutants enter the water body, such as the runoff from an agricultural area. Urban stormwater drainage—which may carry sand and other gritty materials, petroleum residues from automobiles, and road-deicing chemicals—is also considered a dispersed source because of the many locations at which it enters local streams or lakes.

Point sources of water pollution are easier to control than dispersed sources, because the contaminated water has been collected and conveyed to one single point where it can be treated. Pollution from dispersed sources is difficult to control, and, despite much progress in the building of modern sewage treatment plants, dispersed sources continue to cause a large fraction of water pollution problems.

Types of Water Pollution

Water pollution not only endangers the health of humans, it can put entire ecosystems at risk by harming aquatic species. When streams, rivers, lakes, oceans, and other bodies of water become contaminated, it degrades water quality and makes it toxic to humans and the environment. The types of water pollution include:

1. Groundwater Pollution

Many people worldwide rely on groundwater, which is found underground in the spaces between soil and fractured rock. The underground stores of water, or aquifers, are replenished with rainwater.

The ground and seep can absorb pesticides and fertilizers into underwater water sources, compromising the safety of the water, especially for humans to drink. Septic tanks and sewage systems sometimes leak untreated waste into groundwater, leading to groundwater pollution.

2. Surface Water Pollution

Surface water refers to all bodies of water above the Earth's surface, such as oceans, lakes, and rivers. Any hazardous substance that comes into contact with surface water is called surface water pollution.

Surface water pollution occurs in three ways: naturally, accidentally, and intentionally.

- Naturally: Flooding or tsunamis, that pick-up fertilizers, pesticides, debris, and other contaminants.
- Accidentally: Oil spills and agricultural runoff.
- Intentionally: Industries dumping waste directly into waterways.

3. Oil Spillages

Water can be polluted from accidental oil spills, transportation, runoff, and intentionally dumping. Oil spillage can have devastating effects on water environments, and they are extremely difficult to remove; just one liter of oil can pollute one million liters of water.

Oil spills make up 12% of the oil that enters the world's oceans. The rest of the oil spillages come from shipping/transport, draining, and dumping. Most oil spills are localized, but even a small amount of oil can be extremely hazardous to humans and marine life.

Once oil enters the water, it quickly spreads over the surface, reducing the amount of oxygen and sunlight that can penetrate the water's surface preventing plants to photosynthesize, and suffocating fish. Water pollution from oil spillage also smothers the feathers of seabirds, preventing them from flying and exposes them to oil ingestion when they catch fish.

4. Chemical Pollution

The most common culprit for water pollution is chemical pollution. While that includes pesticides and fertilizers, heavy metals and solvents used in industrial sites are major contributors to this pollution.

Contaminants that can find their way into drinking water include:

- arsenic
- fracking fluids
- lead
- chlorinated disinfection products
- PFAS
- uranium
- nitrogen
- bleach

5. Nutrient Pollution

Nutrients like calcium, chloride, fluoride, magnesium, potassium, and sodium are commonly found in our drinking water. However, if the nutrients are too plentiful, they can interfere with an ecosystem, which sometimes does more harm than good.

Water pollutants such as fertilizers can block out sunlight and stunt the growth of underwater organisms, throwing everything out of whack.

6. Microbiological Pollution

Bacteria, viruses, and protozoa contribute to microbiological contamination. The microorganisms responsible for water pollution can be found in human and animal waste, sediments, and underwater aquatic vegetation and can cause life-threatening diseases if not properly decontaminated.

Some developing and low-income countries don't have adequate water treatment systems to filter contaminated water.

7. Suspended Matter Pollution

Suspended matter pollution comes from soil erosion, runoff, discharges, stirred bottom sediments, and algae blooms. Suspended solids interfere with water chemistry and microbiology, negatively impacting human health and the environment.

Suspended particles either settle to the bottom or float to the surface. When they settle, they damage underwater life and leach chemicals into the water supply. Floating particles on the surface can prevent oxygen and sunlight from penetrating the water.

8. Thermal Pollution

Human behavior and natural occurrences can lead to thermal pollution, which is the degradation of water quality due to changes in water temperature.

9. Oxygen Depletion Pollution

Algal blooms, a form of water contamination, profoundly impact aquatic ecosystems and water quality. These blooms occur when certain algae rapidly reproduce, forming dense concentrations in water bodies. Their growth leads to high oxygen consumption, causing oxygen depletion harmful to other aquatic life forms.

Additionally, algal blooms worsen water pollution by releasing toxins. Some algae species produce toxins, which are released in significant amounts during blooms. These toxins endanger human and animal health, particularly when ingested through contaminated water. They can cause various health issues, including gastrointestinal problems, liver damage, and neurological effects. Controlling nutrient pollution, a primary cause of algal blooms, is crucial for preserving water quality and safeguarding public health.

****Water pollution can also be classified into physical, chemical, and biological pollution based on the type of contaminants present in the water.

1. Physical Water Pollution

Physical pollution changes the physical properties of water, such as its appearance, temperature, or clarity.

Examples:

- Suspended soil and sediments
- Plastic waste and litter
- Industrial solid waste
- Thermal pollution (hot water discharged from power plants or industries)

Effects:

- Reduces water transparency
- Blocks sunlight needed by aquatic plants
- Increases water temperature, reducing dissolved oxygen
- Harms aquatic habitats

2. Chemical Water Pollution

Chemical pollution occurs when harmful chemicals enter water bodies.

Examples:

- Pesticides and herbicides from agriculture
- Fertilizers containing nitrates and phosphates
- Heavy metals (lead, mercury, cadmium)
- Industrial chemicals and acids
- Oil spills and petroleum products
- Detergents and household chemicals

Effects:

- Toxic to aquatic organisms
- Can contaminate drinking water
- Causes health problems in humans, such as poisoning and cancer
- Leads to eutrophication (excessive growth of algae)

3. Biological Water Pollution

Biological pollution is caused by living organisms or biological materials that contaminate water.

Examples:

- Bacteria (such as *E. coli*)
- Viruses
- Protozoa
- Parasitic worms
- Sewage and animal waste
- Harmful algal blooms

Effects:

- Spreads waterborne diseases such as cholera, typhoid, and dysentery
- Makes water unsafe for drinking and recreation
- Can reduce oxygen levels and harm aquatic life

Effect of water pollution on land

Water pollution directly degrades adjacent land by contaminating soil, reducing agricultural yields, and accumulating toxic heavy metals in the food chain. Polluted irrigation water introduces harmful pathogens and chemicals into the earth, threatening local ecosystems and human health.

Specific effects of water pollution on land include:

- **Soil Contamination & Fertility Loss:** Toxic industrial wastewater and chemical runoff often seep into the ground, altering the soil's chemistry and destroying essential microorganisms. This degradation stunts plant growth and reduces overall soil fertility.
- **Decreased Agricultural Productivity:** Farmers relying on contaminated rivers or lakes for irrigation frequently suffer from poor crop yields and compromised crop quality.
- **Food Chain Disruption:** Hazardous substances like lead and cadmium can transfer from polluted water to crops, livestock, and eventually humans, causing long-term health risks through biomagnification.
- **Vegetation Destruction:** Acidic or oil-heavy water runoff can poison riparian vegetation, stripping away natural ground cover and accelerating topsoil erosion.
- **Wastewater Risks:** Using untreated wastewater in livestock farming can introduce infectious pathogens into both the soil and the food supply.

Effect of water pollution on ground water

Water pollution profoundly threatens groundwater aquifers by allowing hazardous surface contaminants—such as heavy metals, pathogens, and agricultural chemicals—to seep into underground water sources. Once polluted, groundwater is incredibly difficult and costly to clean, creating severe public health, ecological, and socioeconomic crises.

Key Effects on Public Health

- **Waterborne Diseases:** Consuming pathogen-laden groundwater leads to acute illnesses like cholera, dysentery, and hepatitis.
- **Chronic Toxicity:** Heavy metals (e.g., arsenic, lead) and industrial chemicals seep into aquifers, elevating the risk of cancers, nerve damage, and developmental issues. Arsenic contamination is a primary concern, directly impacting millions worldwide, particularly in densely populated delta regions like Bangladesh.
- **Blue Baby Syndrome:** High levels of nitrates (typically from agricultural fertilizers) in drinking water cause methemoglobinemia, a condition that severely impacts infants.

Ecological & Agricultural Damage

- **Soil Degradation:** When polluted groundwater is pumped for irrigation, toxic compounds accumulate in the soil, stunting crop growth and reducing food safety.

- **Disrupted Ecosystems:** Contaminated groundwater eventually discharges into lakes, wetlands, and rivers, introducing persistent toxins that threaten aquatic biodiversity and alter local ecosystems.

Socioeconomic Repercussions

- **Water Scarcity:** As water tables drop due to over-extraction, the concentration of pollutants increases. Entire communities or industrial sectors may be forced to find alternative, expensive sources of clean water.
- **Costly Remediation:** Filtering and treating polluted aquifers require extensive, long-term interventions that place a massive financial burden on local governments

Effect of water pollution on surface water

Water pollution degrades surface water by reducing water quality, disrupting aquatic ecosystems, and threatening public health. Pollutants like toxic chemicals, raw sewage, and agricultural runoff trigger severe ecological disruptions and pose direct risks to humans and wildlife.

Surface water pollution triggers several cascading effects on our ecosystems and communities:

1. Ecosystem and Wildlife Degradation

- **Eutrophication:** Nutrient pollution (e.g., nitrogen and phosphorus from fertilizers) triggers massive, toxic algal blooms. These blooms block sunlight and consume dissolved oxygen as they decompose, suffocating aquatic life and creating aquatic "dead zones".
- **Loss of Biodiversity:** Toxic chemicals and heavy metals entering waterways can poison fish, amphibians, and birds. This disrupts the entire food chain, causing population declines or local extinctions of certain species.
- **Bioaccumulation:** Heavy metals (like mercury) and microplastics accumulate in the bodies of smaller organisms and become highly concentrated as they move up the food chain to apex predators and humans.

2. Severe Human Health Risks

- **Waterborne Diseases:** When surface waters (such as rivers and lakes) are contaminated with untreated domestic sewage or animal waste, they carry pathogenic bacteria, viruses, and parasites. Ingestion of this water causes dangerous diseases like typhoid, cholera, and hepatitis.
- **Chemical Poisoning:** Industrial wastewater and mining runoff introduce toxic chemicals and heavy metals. Exposure to these pollutants can lead to acute poisoning, chronic organ damage, and an increased risk of cancer.

3. Agricultural and Economic Repercussions

- **Contaminated Crops:** Using polluted surface water to irrigate farmlands can introduce toxins into crops, reducing agricultural yields and endangering food safety.
- **Economic Strain:** The cost to decontaminate and purify heavily polluted surface waters for drinking and commercial use requires massive investments in municipal water treatment.

Effect of water pollution on aquatic life

Water pollution devastates aquatic ecosystems by introducing toxic chemicals, depleting dissolved oxygen, and altering water chemistry. These contaminants cause rapid die-offs, disrupt reproductive systems, and poison food chains, ultimately leading to significant losses in aquatic biodiversity and habitat destruction.

The most critical and far-reaching effects on aquatic life include:

- **Oxygen Depletion & Dead Zones:** Runoff rich in nitrogen and phosphorus (from agricultural fertilizers) triggers rapid algae growth. When these algae die and decompose, they consume vast amounts of dissolved oxygen, creating hypoxic "dead zones" where fish and invertebrates suffocate.
- **Toxicity and Bioaccumulation:** Heavy metals (like mercury) and pesticides enter water systems and accumulate in the bodies of aquatic organisms. Over time, these toxins magnify as they move up the food chain, severely impairing the neural functions, immune systems, and reproductive success of predators.
- **Physical Harm & Habitat Damage:** Oil spills physically coat marine animals, destroying the insulation of birds and mammals, while petroleum toxins cause gill damage and fin rot in fish. Furthermore, marine plastic debris causes lethal entanglement and fatal obstructions in the digestive tracts of fish, sea turtles, and dolphins.
- **Disrupted Chemical Balances:** Pollutants alter the fundamental properties of water, such as its pH and temperature (thermal pollution). This makes the environment hostile to native flora and fauna, often prompting the spread of invasive, opportunistic species.

Effect of water pollution on sea

Water pollution causes widespread, systemic damage to marine ecosystems, with up to 80% of sea pollution originating from land-based run-off and sewage. It destroys habitats, contaminates food webs, depletes oxygen in the water, and introduces toxic chemicals that threaten both aquatic life and human health.

Here is a breakdown of how different types of pollution impact the sea:

1. Plastic & Microplastics

- Ingestion and Entanglement: Over 100,000 marine mammals and countless fish die annually from ingesting plastics or becoming entangled in discarded fishing gear.
- Microplastics: Plastics break down into microscopic particles that are consumed by small organisms. These toxins then accumulate up the food chain, a process that poses direct risks to humans who consume seafood.

2. Nutrient Pollution (Eutrophication)

- Algal Blooms: Run-off carrying agricultural fertilizers introduces excess nitrogen and phosphorus, which trigger massive, explosive growths of algae.
- Dead Zones: As the algae die and decompose, they consume massive amounts of oxygen in the water. This creates vast, suffocating "dead zones" where marine life cannot survive.

3. Oil Spills

- Coating Wildlife: Oil spills form thick surface slicks that coat seabirds' feathers and marine mammals' fur, stripping away their natural insulation and ability to swim or fly.
- Toxicity: Ingesting crude oil poisons marine life and can cause severe reproductive problems, gill damage, and immediate mass die-offs.

4. Chemical & Wastewater Pollution

- Pathogens and Coral Bleaching: Untreated wastewater and industrial chemicals introduce viruses and harmful bacteria into the water, causing diseases (such as white pox) in coral reefs and shellfish.
- Heavy Metals: Toxins like mercury and pesticides poison marine life at all trophic levels, hindering ocean recovery and changing food web structures.

Ecological System and Water Pollution

An **ecosystem** is a community of living organisms (plants, animals, and microorganisms) interacting with each other and with their physical environment, including water, air, and soil. Water pollution can seriously disrupt the balance of ecosystems.

Effects of Water Pollution on Ecosystems

1. Loss of Biodiversity

- Pollutants can kill sensitive species of plants and animals.
- Some species may become endangered or disappear from affected areas.

2. Disruption of Food Chains

- Toxic substances can accumulate in organisms and move through the food chain.
- Predators at higher levels often receive the highest concentrations of pollutants.

3. Eutrophication

- Excess nutrients from fertilizers cause rapid growth of algae.
- Algal blooms block sunlight and reduce oxygen levels when they decompose.
- Fish and other aquatic organisms may die due to oxygen depletion.

4. Habitat Destruction

- Pollution damages aquatic habitats such as wetlands, rivers, lakes, and coral reefs.
- Organisms lose breeding, feeding, and shelter areas.

5. Reduced Water Quality

- Contaminated water becomes unsuitable for many plants and animals.
- Changes in pH, temperature, and chemical composition can stress or kill organisms.

6. Bioaccumulation and Biomagnification

- Toxic chemicals such as mercury and pesticides accumulate in living organisms (**bioaccumulation**).
- Their concentration increases at higher levels of the food chain (**biomagnification**).

7. Effects on Ecosystem Services

- Healthy ecosystems provide clean water, food, flood control, and climate regulation.
- Water pollution reduces the ability of ecosystems to provide these services.

Example

When fertilizers enter a lake:

1. Algae grow rapidly.
2. Oxygen levels decrease.
3. Fish die from lack of oxygen.
4. Birds and other animals that depend on fish lose a food source.
5. The entire ecosystem becomes imbalanced.

Water pollution disrupts ecosystems by harming organisms, destroying habitats, reducing biodiversity, and interfering with natural ecological processes. Protecting water quality is essential for maintaining healthy and balanced ecosystems.

Principal Industries Responsible for Water Pollution

Several industries contribute significantly to water pollution by discharging untreated or inadequately treated wastewater into rivers, lakes, and groundwater.

1. Textile Industry

- Releases dyes, bleaching agents, detergents, and chemicals.
- Causes discoloration of water and chemical contamination.

2. Chemical Industry

- Discharges acids, alkalis, solvents, and toxic chemicals.
- Can severely affect aquatic ecosystems and human health.

3. Pulp and Paper Industry

- Produces wastewater containing organic matter, chlorine compounds, and suspended solids.
- Reduces oxygen levels in water bodies.

4. Leather and Tannery Industry

- Releases chromium, sulfides, and other toxic chemicals.
- Can contaminate both surface water and groundwater.

5. Food Processing Industry

- Produces wastewater rich in organic waste, oils, and fats.
- Leads to oxygen depletion when discharged into water bodies.

6. Petroleum and Oil Refining Industry

- Releases oil, hydrocarbons, and toxic chemicals.
- Causes water contamination and harms aquatic life.

7. Mining Industry

- Produces wastewater containing heavy metals and acidic drainage.
- Pollutes rivers, lakes, and groundwater sources.

8. Pharmaceutical Industry

- Discharges chemicals, antibiotics, and drug residues.
- Can affect aquatic organisms and contribute to antimicrobial resistance.

9. Metal Processing and Electroplating Industry

- Releases heavy metals such as lead, cadmium, nickel, and chromium.
- These metals are toxic and can accumulate in the food chain.

10. Power Plants

- Discharge heated water used for cooling processes.
- Cause thermal pollution, reducing dissolved oxygen levels in water.